> who was not prepared to make the least concession or grant that he could ever be mistaken. It is hardly possible to convey just how passionate the discussions were, just how deeply rooted the convictions of each, a fact that marked their every utterance.

Schrödinger believed that his wave equation's success meant that all quantum phenomena could eventually be explained away as the behavior of continuous waves. Yet Bohr and Heisenberg pointed out that there were phenomena that seemed to demand quantum "jumps," like electrons jumping from one orbit to another in Bohr's atom, which could not be explained away by the smooth movement of waves. Schrödinger disagreed. "If all this damned quantum jumping were really here to stay, I should be sorry I ever got involved with quantum theory," he complained. Eventually, Schrödinger, worn out by Bohr's relentless questioning, caught a "feverish cold" in the damp dark Danish autumn and took to his bed in the Bohrs' house. While Margrethe, Bohr's wife, brought Schrödinger tea and cake, Bohr continued to press his advantage, sitting on the edge of Schrödinger's bed and saying in his quiet voice, "But you must surely admit that.…"

With neither side convinced by the other, Schrödinger went home. "No understanding could be expected, since, at the time, neither side was able to offer a complete and coherent interpretation of quantum mechanics," Heisenberg recalled. "For all that, we in Copenhagen felt convinced toward the end of Schrödinger's visit that we were on the right track." Fundamentally, the problem was that the meaning of Schrödinger's wave function was still not clear. But Max Born had discovered a piece of the puzzle that summer. He found that a particle's wave function in a location yields the probability of measuring the particle in that location—and that the wave function collapses once measurement happens. Born's insight ultimately won him a Nobel Prize, and rightly so. But Born's rule for handling wave functions also left physicists with new puzzles: What was a measurement? And why did wave functions behave differently when they were being measured, whatever that might mean? Born's idea and Schrödinger's mathematics had unlocked the quantum world, but at a price. The measurement problem had arrived.